
Matrix-Free Covariance-Response Approach for Data Assimilation-Ready Weather Ensembles

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Abstract

We propose a lightweight, matrix-free covariance-response evaluation to assess data assimilation (DA) readiness before expensive complete cycling. Given ensemble perturbations and a localized observation-like probe, we compute *covariance actions* without forming dense covariance matrices. On NOAA/NCEP Global Ensemble Forecast System (GEFS) 500 hPa geopotential height (HGT500) data, a query-conditioned, analog-style historical response improves localized agreement with the query-time covariance response while preserving matrix-free scalability.

Motivation and framework. Weather and climate prediction are based on high-dimensional, computationally expensive physics-based models. Ensembles, or Monte Carlo forecasts, quantify uncertainty by representing plausible forecast-state perturbations. DA combines observations with background forecast estimates to provide improved initial conditions for subsequent forecasts. Recent AI emulators can reduce the forecast cost, but their ensembles must be evaluated not only for state accuracy or spread, but also for whether their perturbations produce DA-relevant covariance structure. This motivates an observation-centered covariance-response evaluation. In DA, a localized observation should influence nearby and dynamically related state variables through the localized

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background-error covariance. Therefore, assimilating a localized observation should produce an increment pattern whose shape reflects the covariance structure of the background ensemble.

Let $x_{q,1}, \dots, x_{q,N_e} \in \mathbb{R}^K$ be the ensemble members at query time q , and let $\bar{x}_q = N_e^{-1} \sum_{i=1}^{N_e} x_{q,i}$. With the perturbation matrix $\mathbf{X}_q = [x_{q,1} - \bar{x}_q, \dots, x_{q,N_e} - \bar{x}_q]$, the empirical covariance is $\mathbf{B}_q = (N_e - 1)^{-1} \mathbf{X}_q \mathbf{X}_q^\top$. We compute its action $\mathbf{B}_q \mathbf{v} = (N_e - 1)^{-1} \mathbf{X}_q (\mathbf{X}_q^\top \mathbf{v})$, for a localized probe $\mathbf{v} \in \mathbb{R}^K$ replacing dense $\mathcal{O}(K^2)$ covariance storage/application by $\mathcal{O}(K \cdot N_e)$ cost and memory. The localization remains matrix-free by restricting to a local window or applying $\mathbf{B}_{\text{loc}} \mathbf{v} = r_{i_0} \odot (\mathbf{B} \mathbf{v})$, where $r_{i_0} \in [0, 1]^K$ is a taper centered on the probe grid index. Using GEFS ensemble data [1], we compare $\mathbf{B}_q \mathbf{v}$ with two historical constructions. The static response $\mathbf{B}_{\text{static}} \mathbf{v} = M^{-1} \sum_{m=1}^M \mathbf{B}_m \mathbf{v}$ averages the historical covariance actions equally. The query-conditioned response is analog-style:

$$\mathbf{B}_{\text{cond}}(q) \mathbf{v} = \sum_{m \in \mathcal{H}(q)} w_m(q) \mathbf{B}_m \mathbf{v}, \quad \text{where} \quad w_m(q) = \frac{\exp(-d(\psi_q, \psi_m)/\ell)}{\sum_{j \in \mathcal{H}(q)} \exp(-d(\psi_q, \psi_j)/\ell)}. \quad (1)$$

Here, $\psi_q, \psi_m \in \mathbb{R}^p$ are local meteorological descriptors, $d(\cdot, \cdot)$ is descriptor distance, $\ell > 0$ is a finite flow-space length scale, and $\mathcal{H}(q)$ is the admissible historical pool.

GEFS result. We evaluate localized responses for NOAA/NCEP GEFS HGT500, i.e., 500 hPa geopotential height, a standard mid-tropospheric variable indicating large-scale weather-system position. For each query, $\mathbf{B}_q \mathbf{v}$ is the reference response. In representative HGT500 cases as demonstrated in Figure 1, the (1) is closer to $\mathbf{B}_q \mathbf{v}$ than the static archive response and assigns a greater influence to meteorologically similar historical cases. This provides a resource-aware, DA-facing screen for learned, archive-based, or generative ensembles before costly full DA cycling.

References

- [1] Hong Guan, Yuejian Zhu, Eric Sinsky, Bing Fu, Wei Li, Xiaqiong Zhou, Xianwu Xue, Dingchen Hou, Jiayi Peng, MM Nageswararao, et al. GEFSv12 Reforecast Dataset for Supporting Subseasonal and Hydrometeorological Applications. *Monthly Weather Review*, 150(3):647–665, 2022. [2](#)

A GEFS HGT500 Result

For each query case, the GEFS response in query-time $\mathbf{B}_q \mathbf{v}$ is treated as the reference response. We compare it with the static archive response and the query-conditioned response. We visualize the response fields and the localized error reduction,

$$\Delta e(\mathbf{v}; q) := |\mathbf{B}_{\text{static}} \mathbf{v} - \mathbf{B}_q \mathbf{v}| - |\mathbf{B}_{\text{cond}}(q) \mathbf{v} - \mathbf{B}_q \mathbf{v}|.$$

Hence, $\Delta e(\mathbf{v}; q) > 0$ indicates locations where the query-conditioned response is closer to the query-time GEFS response than the static baseline. Figure 1 shows the representative flow-conditioning reduces localized response error and assigns greater influence to similar historical cases.

GEFS HGT500 observation-centered covariance response

2026-06-08 00 UTC, lead +6 h | probe: 54.5°N, 138.5°E | local gain = 1.212

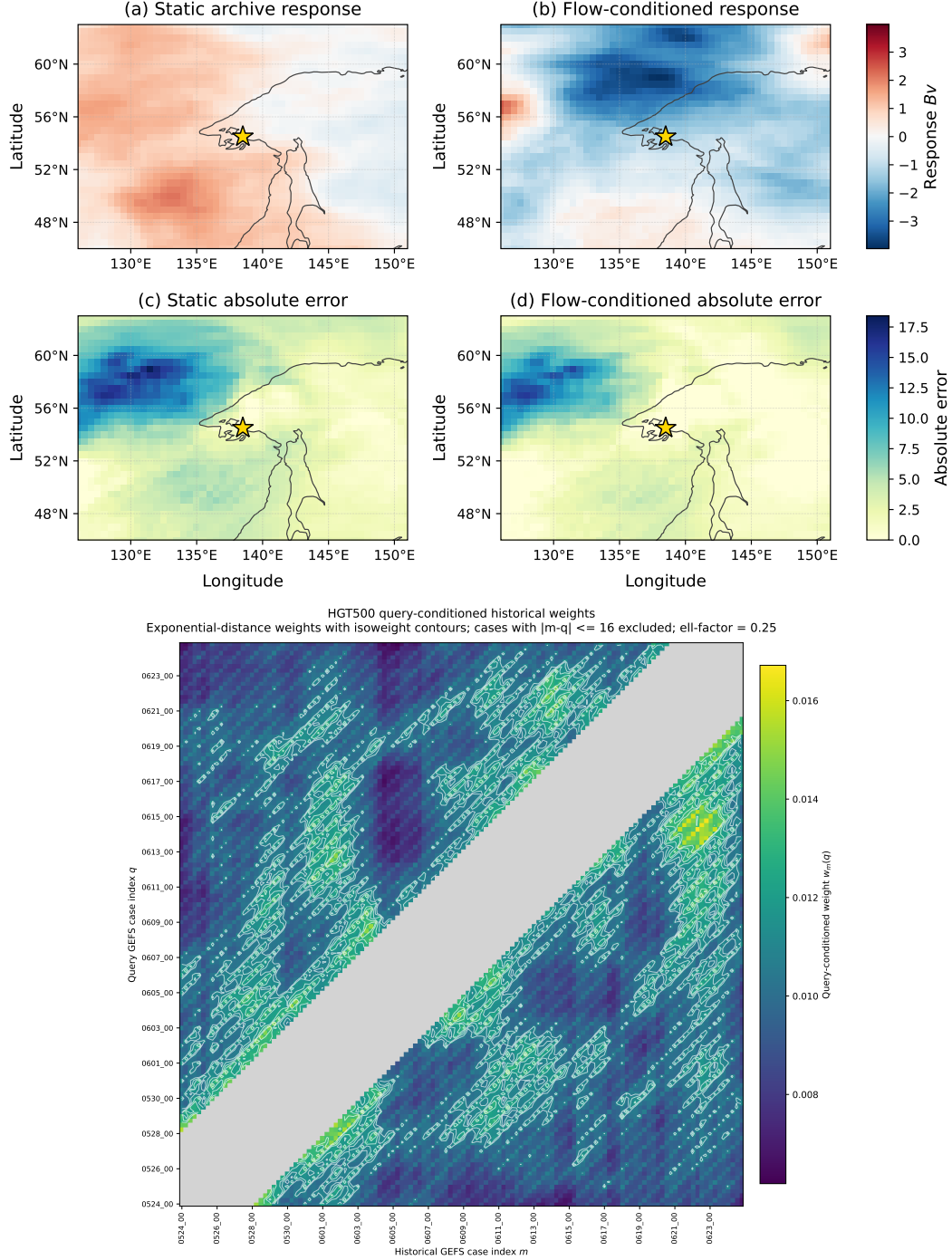


Figure 1: **Matrix-free flow-conditioned covariance response on GEFS HGT500.** *Top:* For the 8 June 2026 00 UTC GEFS case at +6 h lead, we compare static and flow-conditioned covariance responses around a localized probe at 54.5°N, 138.5°E. Top panels show Bv ; bottom panels show absolute errors relative to the query-time GEFS response $B_q v$. The star marks the probe, and the local gain is 1.212. *Bottom:* Query-conditioned historical weights $w_m(q)$ over 24 May–23 June 2026 GEFS cases; the gray band marks near-time cases excluded from the analog pool.